

The Normative-Material Corridor: Governing Maternal Nutrition Through WHO Standards, Philanthropic Capital, and Indonesia's Adoption of Multiple Micronutrient Supplements

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ABSTRACT

This article examines this phenomenon through the global transition from iron-folic acid supplementation to antenatal multiple micronutrient supplements (MMS), using Indonesia as a critical contemporary case. It introduces the concept of a normative-material corridor: a governance structure in which authoritative scientific standards, evidence hierarchies, financing mechanisms, product availability, procurement support, technical assistance, and monitoring processes collectively narrow the range of practically available national policy choices. The corridor does not eliminate state sovereignty or legally mandate adoption. Instead, it makes one policy option progressively more legitimate, feasible, and administratively actionable than its alternatives. Using doctrinal analysis and socio-legal document analysis, this article traces the development of the MMS standard, the translation of World Health Organization authority into policy influence, the expanding role of philanthropic and catalytic capital, and the domestic implementation of these forces through Indonesian health, administrative, budgetary, and procurement frameworks. The analysis demonstrates that legal responsibility remains concentrated within the state, even when agenda setting, evidence generation, and material capacity are distributed among transnational actors. Indonesia therefore appears neither as a passive policy recipient nor as an entirely unconstrained sovereign actor, but rather as an agent exercising structured agency within a policy field whose prevailing conditions have already been shaped. To address the resulting accountability gap, this article proposes six safeguards: domestic legal translation, public reason-giving, financial and relational transparency, adaptive review mechanisms, a systems obligation against the reduction of stunting management to tablet-based interventions, and a distributed accountability matrix. MMS consequently reveals a broader challenge for medical law: how to govern forms of power that are non-coercive in appearance but regulatory in effect.

INTRODUCTION

Maternal nutrition is often framed as a clinical problem: identify deficiency, provide supplementation, and measure biological responses. Although this approach is necessary, it remains incomplete. Micronutrient deficiencies are embedded within broader systems involving food availability, poverty, gendered resource allocation, access to antenatal care, infectious diseases, pharmaceutical supply chains, and public administration (Stevens et al., 2022; Passarelli et al., 2024). Their consequences extend across maternal morbidity, fetal growth, gestational duration, birth weight, and neonatal survival. A legal analysis must therefore examine not only whether an intervention is effective, but also who defines the intervention, how it becomes a standard, which institutions make it feasible, and where responsibility lies when a transnational policy is implemented through domestic systems (Sekalala & Masud, 2021; Ralston et al., 2024). These questions frame the central inquiry of

this article: how transnational governance shapes national maternal nutrition policy without relying on binding international law (Biswas, 2021).

The intergenerational implications make this inquiry particularly important. Conditions during gestation can influence childhood growth, long-term health, educational attainment, and economic participation. Maternal nutrition is therefore not merely an element of prenatal care; it is a domain in which present distributive decisions influence future capabilities. Although the language of human capital often captures the magnitude of these effects, it may also obscure women as rights-bearing individuals by framing their bodies primarily as instruments for producing healthier children (Paiero & de Moraes, 2027).

Anaemia illustrates the limitations of a single-variable approach. It remains prevalent among women, yet its occurrence reflects multiple factors, including iron deficiency, other micronutrient deficiencies, infection, inflammation, haemoglobin disorders, and social conditions (GBD 2021 Anaemia Collaborators, 2023). A policy that equates maternal nutrition solely with iron supplementation may be administratively convenient while remaining biologically and socially incomplete (Benson et al., 2021; Turawa et al., 2021). The same caution applies to stunting. Comparative evidence identifies multiple prenatal and postnatal determinants, including maternal characteristics, fetal growth restriction, infection, dietary inadequacy, water and sanitation conditions, and socioeconomic factors (Beal et al., 2018; Vaivada et al., 2020). Height-for-age is consequently a population-level indicator of cumulative deprivation rather than a diagnosis that identifies a single causal pathway or a single corrective intervention (Vaivada et al., 2020).

Against this background, multiple micronutrient supplements (MMS) have moved from being an intervention under investigation to an increasingly prominent component of global maternal nutrition policy. This transition raises a legal puzzle. The World Health Organization (WHO) does not generally compel states to procure particular antenatal supplements through binding legal mechanisms. Philanthropic foundations, United Nations agencies, and technical consortia do not possess general legislative authority over national health systems. Nevertheless, their standards, financing mechanisms, product specifications, market-shaping activities, and implementation support can substantially influence the range of realistic domestic policy choices (Mazzucato et al., 2020; Mazzucato & Kattel, 2026; Mazzucato & Li, 2021; Mazzucato & Penna, 2016; Wagner et al., 2026).

This article argues that the binary distinction between binding and non-binding law is insufficient to explain this form of influence. It introduces the concept of the normative-material corridor: a governance structure in which scientific authority and material enablement operate together. Normative authority identifies a preferred intervention, establishes the evidentiary framework through which it is justified, and provides a vocabulary of international legitimacy. Material enablement makes that intervention available through financing, product supply, technical assistance, procurement templates, data systems, and implementation partnerships. Neither element alone determines national policy. Together, however, they can make one course of action easier to justify, finance, procure, and scale than its alternatives. The concept therefore provides an analytical framework for understanding how non-coercive forms of power can generate regulatory effects.

Indonesia provides a critical case for examining this dynamic. It is a large middle-income democracy with a decentralized health system, substantial domestic administrative

and industrial capacity, persistent challenges related to maternal anaemia and child undernutrition, and an established constitutional and statutory framework for health. It is neither a weak state passively receiving externally designed programs nor a state operating independently of transnational governance. Indonesia's adoption and scaling of MMS demonstrate a form of structured agency: the state selects, co-finances, and adapts policy, but does so within an environment in which evidentiary standards, product configurations, and mechanisms for policy acceleration have already been shaped by global actors. Indonesia therefore provides an instructive case for examining how domestic law mediates transnational governance.

This article makes four contributions. First, it distinguishes soft governance from soft law and identifies the interaction between normative and material forms of power. Second, it explains why formal state consent alone does not fully resolve legitimacy concerns when the practical conditions of policy choice are externally structured. Third, it provides a contemporary legal analysis of Indonesia's MMS scale-up. Fourth, it develops safeguards that preserve the speed and potential health benefits of transnational cooperation while strengthening public reasoning, transparency, reversibility, and rights protection. Sections II to V develop the conceptual and institutional argument; Sections VI to VIII apply it to the governance architecture and Indonesian implementation; Section IX develops legal and ethical safeguards; and Section X concludes. By integrating these contributions, this article aims to advance theoretical understanding of global health governance and contribute to the practical governance of maternal nutrition policy in Indonesia.

METHOD

The article combined doctrinal analysis with socio-legal document analysis. The doctrinal component examined the legal status and accountability implications of WHO recommendations, international health and human rights instruments, and Indonesian constitutional, health, administrative, transparency, public service, financial, and procurement laws. The socio-legal component examined how evidence, standards, funding, and implementation arrangements interacted across institutional levels. The analysis was based on publicly available materials available up to 16 July 2026. It did not seek to measure program impact, infer the contents of unpublished donor agreements, or substitute documentary analysis for empirical research involving pregnant women, health workers, procurement officials, or local manufacturers.

Document analysis was appropriate for addressing the legal question because relevant forms of power were often embedded in institutional instruments and governance relationships rather than expressed through direct commands. However, documents also represented curated expressions of institutional objectives. Guidelines emphasized evidentiary approaches, press releases highlighted institutional commitments, and legislation established formal authority. Therefore, this article treated official descriptions as evidence of declared governance structures rather than proof of implementation outcomes. It also distinguished three propositions that are often conflated: that MMS may generate beneficial health outcomes; that national adoption may be a reasonable policy choice; and that the governance process through which adoption occurs must remain legally accountable. Acceptance of the first two propositions did not determine the third.

RESULTS AND DISCUSSION

From Ifa To Mms: The Making Of A Global Standard

1. The Administrative Appeal of a Single-Nutrient Default

Iron-folic acid supplementation (IFA) became a durable feature of antenatal policy because it combined a clear clinical rationale with an administratively legible product. Iron addressed a major cause of anaemia, folic acid reduced the risk of neural tube defects, and a standard tablet could be incorporated into routine antenatal contacts. WHO's 2016 antenatal care guideline located nutrition within a broader positive-pregnancy-experience framework, but IFA remained an established default in many national systems. Defaults matter because health administrations do not choose among interventions on efficacy alone. They inherit formularies, training materials, supply contracts, reporting indicators, public expectations and professional routines.

The emergence of the United Nations International Multiple Micronutrient Antenatal Preparation (UNIMMAP) altered the object of governance. Rather than addressing a single deficiency, the formulation assembled 15 vitamins and minerals into a standard product for use in programme and research settings. Standardisation enabled trials to become comparable, manufacturers to work from a common specification, donors to finance a recognisable commodity and ministries to contemplate procurement at scale. A tablet thus became both a biomedical intervention and a regulatory artefact: its composition stabilised expectations across research, markets and public programmes.

2. Evidence, Heterogeneity and the Meaning of Benefit

Systematic review evidence generally indicates that antenatal MMS can reduce low birth weight and small-for-gestational-age births relative to IFA, while effects on other outcomes are smaller, uncertain or context dependent. Individual-participant-data analysis has further suggested that benefits may vary according to maternal anaemia, nutritional status, timing and other characteristics. Such heterogeneity is not a reason to dismiss the intervention. It is a reason to resist translating a population-average estimate into an unconditional legal or administrative mandate.

Earlier syntheses already showed that the evidentiary question was not whether every outcome improved, but whether a bundle of modest benefits justified a change in standard practice. Meta-analysis of birth size and gestational outcomes supported the possibility of population-level gains while also exposing differences among trial settings, formulations and baseline risks. The trial literature itself spans distinct health systems. A community trial in rural Nepal, for example, tested alternative supplement combinations under conditions that cannot be assumed to match routine service delivery elsewhere.

The JiVitA-3 trial in rural Bangladesh added large-scale evidence on birth outcomes and infant mortality, but it also demonstrated the distance between a carefully managed trial and a national programme that must maintain supply, counselling, adherence and referral across diverse facilities. Cost-effectiveness modelling has strengthened the policy case by estimating that small per-pregnancy cost differences can be outweighed by reductions in adverse birth outcomes. Yet modelling necessarily embeds assumptions about prices, coverage, adherence, effect size, service delivery and valuation. A cost-effective intervention in a model becomes a lawful and legitimate public programme only after those assumptions are examined within domestic institutions.

3. The WHO Recommendation as a Policy Signal

WHO's 2020 update did not present MMS as a simple universal command. It recommended antenatal MMS containing iron and folic acid in the context of rigorous research, thereby combining endorsement with a continuing demand for learning. That formulation is legally non-binding and epistemically cautious, but it is also a powerful policy signal. Ministries, donors and advocates can cite the recommendation as evidence that MMS lies within the range of internationally legitimate action. The research condition may then be operationalised broadly through implementation research, monitoring or phased scale-up rather than through conventional randomised trials.

Inclusion of an antenatal multiple micronutrient supplement in the WHO Model List of Essential Medicines further changes the institutional meaning of the product. Essential-medicine status does not compel national listing or procurement. It does, however, strengthen claims of medical necessity, inform model formularies, reduce the reputational risk of adoption and support market forecasting. The movement from formulation, to trial evidence, to guideline, to essential-medicine recognition illustrates how a scientific proposition accumulates normative weight across institutional sites.

Soft Law, Soft Governance And The Normative-Material Corridor

1. Why the Hard-Law/Soft-Law Binary Is Insufficient

International legal scholarship has long described rules as harder or softer according to the degree of obligation, precision and delegation. That framework clarifies why a WHO recommendation differs from a treaty obligation, but it does not fully explain how a non-binding guideline becomes the operational default of a national programme. Soft-law analysis correctly observes that non-binding instruments can influence expectation, interpretation and later lawmaking. It also shows that treaties and soft instruments frequently interact rather than occupy separate worlds.

The deeper issue is that legal softness can coexist with practical firmness. An intervention may be optional in formal doctrine yet difficult to reject once it is endorsed by an authoritative organisation, attached to finance, embedded in procurement specifications, supported by implementation partners and measured through programme indicators. Soft and hard law may function as alternatives, complements or antagonists, but the MMS case adds a material dimension: the cost of choosing outside the endorsed pathway can rise even without a sanction.

Contemporary international rulemaking increasingly occurs through dynamic institutional processes rather than a single act of legislation. Global administrative law responded to this reality by focusing on transparency, participation, reason-giving, review and accountability in transnational administration. Yet health programmes also depend on commodities, logistics and finance. A framework centred only on decision procedure misses the way material capacity selects which norms can travel.

2. Defining Soft Governance

Soft governance, as used here, is the coordinated production of policy effects through non-binding standards, expertise, financing, market organisation, technical assistance, indicators and institutional relationships. It is broader than soft law because not all of its instruments claim normative status. A grant agreement, a product donation, a pooled

procurement arrangement or a digital reporting platform may be legally ordinary, but its placement within a policy architecture can determine which intervention is feasible.

The normative-material corridor has five defining features. First, it is directional: it privileges a pathway without legally eliminating alternatives. Secondly, it is cumulative: the force of the corridor arises from the interaction of multiple instruments. Thirdly, it is distributed: no single actor necessarily controls the whole process. Fourthly, it is reversible in theory but potentially costly to exit in practice. Fifthly, it transfers implementation and legal responsibility to domestic institutions even when important conditions have been shaped elsewhere.

The concept has boundaries. Not every policy diffusion process is a corridor, and not every grant attached to a guideline amounts to problematic influence. A corridor exists only where normative endorsement and material support are sufficiently aligned to alter the comparative feasibility of domestic options. Its presence can be disproved where governments have robustly assessed competing interventions, possess independent financing and supply capacity, retain low-cost exit, and provide meaningful avenues of contestation. The concept is therefore not a rhetorical label for global influence. It is an analytic claim about the architecture and intensity of choice-shaping.

The corridor should not be equated with coercion or conspiracy. It may enable a government to implement an intervention it independently values but cannot yet finance or organise. It may reduce transaction costs, improve quality and accelerate access. Its legal significance lies precisely in that ambivalence. Power can be productive and beneficial while still requiring justification and control.

3. Expertise, Standards and Polycentric Authority

Epistemic communities help explain how shared causal beliefs and professional standards coordinate policy across borders. International organisations add bureaucratic authority: they classify problems, define appropriate knowledge and produce categories that states use to govern. In maternal nutrition, the definition of the relevant deficiency, the choice of outcomes, the formulation of the product and the threshold for recommendation all shape the eventual policy.

Standards are especially influential because they coordinate without appearing to command. They reduce uncertainty for laboratories, manufacturers, procurers, clinicians and evaluators. Private and transnational standard setters can consequently exercise regulatory authority even when national governments retain formal lawmaking power. This authority is polycentric: WHO, UNICEF, expert groups, foundations, researchers, manufacturers, ministries, regulators and service providers each control different parts of the pathway.

Polycentric governance creates a legitimacy problem because influence and responsibility do not align automatically. An actor may shape a standard without funding it, fund a product without implementing it, procure it without counselling users, or implement it without controlling the supply chain. Legitimacy must therefore be constructed and contested across the regime rather than attributed to one sovereign centre. Experimentalist governance offers one response by treating standards as revisable, requiring reporting and allowing local learning to reshape common goals. The challenge is to ensure that experimentation is genuine rather than a label applied to a predetermined scale-up

Who Normative Authority: When Evidence Becomes Regulatory Power

1. Constitutional Authority without Direct Command

The WHO Constitution authorises the organisation to act as a directing and coordinating authority on international health and permits the World Health Assembly to make recommendations to members. Recommendations under this framework are not treaties and generally do not create directly enforceable domestic duties. Their influence instead depends on institutional reputation, expert procedure, repeated use, alignment with other international actors and domestic uptake.

Guideline production converts a heterogeneous evidence base into a decision-oriented statement. WHO's own handbook structures this process through question formulation, evidence review, assessment of benefits and harms, values, resource use, equity, acceptability and feasibility. These are not purely scientific questions. They are mixed judgments about evidence and public values. Their disciplined consideration strengthens legitimacy, but it also shows why a guideline cannot be understood as science speaking without mediation.

2. Evidence Hierarchies and the Politics of Recommendation

The GRADE approach separates confidence in evidence from the strength of a recommendation and thereby makes explicit that policy may depend on more than effect estimates. This is a major advance over informal expert opinion. Still, the conversion from evidence to recommendation is vulnerable to overstatement, especially when strong recommendations are issued despite low confidence in estimated effects. In MMS governance, the wording of the recommendation, the interpretation of the research condition and the treatment of contextual uncertainty become legally relevant because they influence national reason-giving.

Global health priorities are also shaped through knowledge claims and moral narratives. Problems framed as urgent, measurable and solvable through a scalable intervention are more likely to attract political and financial support. Maternal anaemia and adverse birth outcomes fit this pattern, while slower structural reforms in food security, wages, sanitation or gender equality are harder to package as discrete, countable interventions. The resulting asymmetry does not make MMS misguided. It means the visibility of a solution is partly produced by the governance system that can implement it.

3. Independence, Participation and Non-State Engagement

WHO must engage with non-state actors because expertise, finance, manufacturing and implementation capacity are widely distributed. Its Framework of Engagement with Non-State Actors seeks to manage conflicts of interest, reputational risks and undue influence while preserving beneficial cooperation. For MMS, legitimacy depends on more than disclosure of individual conflicts. It also requires attention to structural influence: which questions are funded, which products are standardised, which outcomes are prioritised and which national actors can participate in interpreting the evidence.

A conditional WHO recommendation can therefore perform three regulatory functions. It certifies that adoption is internationally defensible; it provides a common language for donors and ministries; and it relocates disagreement from whether MMS is legitimate to how quickly and under what monitoring conditions it should be implemented. These effects remain compatible with state sovereignty, but they reduce the explanatory value of formal consent alone.

Domestic authorities may properly defer to WHO on matters requiring specialised expertise, especially where duplicating a global evidence review would waste scarce capacity. Deference, however, should be reasoned rather than automatic. The adopting institution must still identify whether the global question matches the domestic question, whether baseline nutritional conditions and service capacity are comparable, whether resource assumptions hold, and whether implementation can satisfy constitutional and statutory duties. The legal error is not reliance on WHO; it is treating reliance as a substitute for domestic judgment. Epistemic authority can justify a rebuttable presumption, but not an irrebuttable policy conclusion.

Philanthropic And Catalytic Capital As Policy Accelerators

1. From Funding Research to Organising a Field

Large philanthropic foundations have become significant actors in global health through grants to research institutions, international organisations, civil society and implementation partnerships. Their influence is not limited to the amount of money supplied. Funding can connect research agendas, advocacy, product development, market forecasting and country programmes. It can also provide continuity across electoral cycles and move faster than public budgeting.

Historical comparison of the Rockefeller and Gates foundations shows how private wealth can shape global health priorities while operating through different institutional styles. The Gates-oriented model has often emphasised measurable interventions, technological solutions and partnerships capable of scaling them. Studies of GAVI demonstrate both the capacity of such arrangements to mobilise resources and the risk that health-system strengthening becomes defined around the requirements of a particular intervention.

Public-private partnerships can combine complementary capacities, but their performance and legitimacy vary with agenda control, transparency, conflicts of interest and accountability. The critical legal question is not whether private funding should be excluded. It is how public authorities should govern dependence on resources that come with product definitions, timelines, indicators or co-financing expectations.

2. Material Enablement and Democratic Asymmetry

Philanthropic power presents a democratic asymmetry. Donors can select problems and strategies without receiving an electoral mandate from the populations affected, while governments remain answerable for results. At the same time, refusing external support may impose immediate opportunity costs on women and children. A legally serious account must avoid two simplifications: treating philanthropy as neutral generosity and treating every donor-supported intervention as illegitimate domination.

The contemporary MMS field illustrates ecosystem building. The Healthy Mothers Healthy Babies Consortium hosts technical advisory work, advocacy resources and a broad membership while acknowledging support from foundations and humanitarian organisations. Such networks can improve coordination and disseminate knowledge. They can also align expert discourse, advocacy messages and funding around a standardised product. Material enablement thus reinforces normative authority: the recommended intervention is not only endorsed but surrounded by institutions capable of making it happen.

3. Matching Funds, Market Making and Exit

Matching finance is especially significant. It frames external support as a catalyst for domestic commitment rather than a substitute for it. This can strengthen sustainability by requiring government investment. It can also create a ratchet: once coverage expands, political and ethical costs make retrenchment difficult even if external finance ends. The legal adequacy of a programme therefore depends on an exit architecture from the beginning, including budget projections, procurement transition, local production strategy, quality assurance and continuity obligations.

Market making is equally consequential. Product specifications, demand forecasts and pooled purchasing can lower prices and stabilise supply. But they may also narrow the field of eligible products, shape domestic industrial policy and create dependence on a small supplier base. Public law must treat these effects as part of health governance, not as technical matters outside legal analysis.

Assessment of donor influence also requires a counterfactual. The relevant comparison is not between a donor-supported programme and a perfectly autonomous public system. It is often between external support and delayed, fragmented or absent implementation. The baseline may already reflect fiscal limits, inherited procurement patterns and unequal research capacity. A government can rationally accept support while still negotiating conditions and protecting future autonomy. The legal objective is therefore not purity from influence, which is impossible, but governance of influence: visibility, justification, proportionality and the capacity to revise the relationship.

The Architecture Of Mms Governance

1. Six Stages of the Corridor

The governance architecture can be described in six stages. First, evidence production defines outcomes and estimates effects. Secondly, normative consolidation converts evidence into a formulation, guideline and essential-medicine status. Thirdly, material enablement supplies grants, matching finance, product availability, technical assistance and market coordination. Fourthly, domestic translation places the intervention within budgets, regulations, formularies, procurement documents and service standards. Fifthly, street-level execution turns the policy into counselling, dispensing, adherence support and data. Sixthly, feedback and exit determine whether evidence from implementation changes the standard and whether domestic institutions can sustain or revise the programme.

The stages are analytically distinct but institutionally interdependent. A scientifically defensible standard may remain inert without products and money. Funding may fail without lawful procurement and trained workers. A national regulation may be symbolically strong but operationally weak if supply is interrupted. Monitoring may generate large datasets while missing women who discontinue supplementation because of side effects, mistrust or inconsistent counselling.

2. Power without Command, Responsibility without Exclusive Control

The corridor distributes control while concentrating legal responsibility. Global actors can define the preferred standard and enable scale, yet the state authorises public expenditure, regulates product quality, sets service obligations and interacts coercively or authoritatively with residents. The state therefore cannot answer a rights claim by pointing to WHO or a

donor. Conversely, accountability will remain incomplete if only the state is examined while the transnational conditions that shaped its decision are treated as background.

This produces a dual legal demand. Domestic institutions must comply with constitutional and administrative duties when adopting and implementing MMS. Transnational institutions must satisfy legitimacy standards appropriate to the influence they exercise, including transparency, reason-giving, conflict management, participation and responsiveness. The relevant question is not whether every influential actor becomes a state-like duty bearer. It is whether accountability follows function closely enough to prevent zones of power without answerability.

3. A Test for Corridor Intensity

The intensity of a normative-material corridor can be assessed through five questions. How specific and authoritative is the global standard? How dependent is implementation on external finance, commodities or technical capacity? How completely has the standard been translated into domestic law and routine administration? How costly would it be to pause, revise or exit? And how accessible are mechanisms through which affected persons, professionals and local institutions can contest the programme? A corridor is strongest where normative specificity and material dependence are high, domestic alternatives are weak, exit is costly and contestability is low.

This intensity assessment has consequences for legal form. A low-intensity corridor may be adequately governed through ordinary clinical guidance and routine procurement. A high-intensity corridor calls for a more visible domestic decision, stronger disclosure, independent evaluation and a defined transition plan. The degree of proceduralisation should therefore track the practical regulatory effect rather than the formal label placed on the international instrument. A recommendation joined to nationwide matching finance and a standardised supply chain deserves more public-law scrutiny than an advisory document that leaves domestic resources and alternatives unchanged.

Indonesia: Structured Agency In A Transnational Policy Field

1. Constitutional and Statutory Location

Indonesia's Constitution recognises a right to obtain health services and assigns the state responsibility for adequate health-service facilities. Those provisions do not prescribe MMS. They establish the public-law setting in which a supplementation programme must be justified: maternal nutrition policy is an exercise of state responsibility, not a charitable concession.

Law No 17 of 2023 on Health provides an integrated framework covering rights and duties, central and regional governmental responsibility, health efforts, facilities, health resources, pharmaceutical resilience, health technology, information systems, financing, coordination, public participation and oversight. This breadth matters because MMS touches each of those domains. It is simultaneously a preventive health intervention, a pharmaceutical or health-supply product, a procurement object, a data point, a budget commitment and a service delivered through decentralised institutions.

Government Regulation No 28 of 2024 operationalises the Health Law and supplies a further regulatory basis for maternal, child and nutritional health measures. Presidential Regulation No 72 of 2021 frames stunting reduction as a holistic, integrated and quality-oriented national strategy involving adolescents, prospective brides and grooms, pregnant and

breastfeeding women, and children under five, with coordination, monitoring, evaluation, reporting and funding. This framework is legally important because it resists reducing stunting to one commodity. MMS must fit within, rather than displace, the multi-sectoral strategy.

The maternal-service pathway is further structured by Minister of Health Regulation No 21 of 2021, which addresses care before pregnancy, during pregnancy, childbirth and the postpartum period, together with contraception and sexual health services. Adoption of MMS therefore occurs inside an existing continuum of care. Its legal quality depends on whether it strengthens that continuum or becomes a stand-alone distribution target detached from examination, counselling, referral and informed choice.

2. The 2026 National Programme

On 5 March 2026, the Ministry of Health and UNICEF announced a five-year national programme under the Child Nutrition Fund. The programme is intended to operate across all 38 provinces, includes MMS within an expanded maternal nutrition package, aims to strengthen supply chains and digital nutrition information systems, supports local production, aligns with the 2025-29 national development plan, and anticipates that more than 4.3 million pregnant women will benefit by 2029. The Fund is to match resources allocated by the Government of Indonesia. These features make the programme an unusually clear example of the normative-material corridor.

The programme should not be characterised as external imposition. Indonesia has articulated its own nutrition priorities, contributes public funds, convenes multiple ministries and agencies, and seeks local production. Those acts constitute meaningful agency. Yet the product standard, international evidence narrative, catalytic finance and implementation ecosystem are not domestically generated in isolation. Agency is exercised within a structured field.

Decentralisation complicates that agency. A national decision must travel through provincial and district planning, health offices, primary-care facilities, logistics systems and local political priorities. Central authorities may define the standard and finance part of the programme, while local governments bear operational burdens and encounter heterogeneous needs. The corridor is therefore vertical as well as transnational: global authority shapes national policy, and national policy structures local discretion. Legitimacy requires upward and downward accountability. Local institutions should report implementation constraints to the centre, but they should also receive reasons, resources and genuine capacity to adapt rather than merely instructions and targets.

3. Administrative Legality and Public Reason

Indonesian administrative law requires governmental action to rest on lawful authority, proper procedure and general principles of good governance. For MMS, legality should be understood substantively. The adopting authority should identify the evidence relied on, explain the treatment of WHO's research condition, assess alternatives, disclose material implementation assumptions and specify review arrangements. A bare reference to international best practice is not enough because it substitutes external authority for domestic reason.

Public information law supports disclosure of policy rationales, agreements, budgets, procurement information and monitoring results, subject to lawful exceptions. Transparency is particularly important where donor support, technical advice and public procurement

intersect. Disclosure should extend beyond the headline value of a grant to the architecture of influence: product specifications, co-financing conditions, indicators, intellectual contributions, data access, branding, exit terms and the role of implementing partners.

Public service law adds a user-facing dimension. Women are not merely beneficiaries counted against a coverage target; they are service users entitled to standards, information, complaint channels and accountable delivery. The relevant quality indicators therefore include consistent availability, understandable counselling, respectful treatment, management of side effects, ability to ask questions, and continuity when a woman moves between facilities or regions.

4. Budget, Procurement and Industrial Policy

State-finance law requires public expenditure to be authorised, accountable and directed to public purposes. Matching funds do not relax those requirements. They make multi-year fiscal analysis more urgent. Government must know the full recurrent cost after external support, including procurement, warehousing, distribution, training, information systems, supervision, pharmacovigilance or product-quality monitoring, and communication.

Government procurement rules provide the legal channel through which a global product standard becomes a domestic commodity. Specifications should protect quality without unjustifiably foreclosing competition or domestic capability. Support for local production may improve resilience and bargaining power, but it must be accompanied by transparent qualification, regulatory oversight and realistic demand forecasting. Procurement decisions also need contingency plans for shortages, supplier failure and changes in the recommended formulation.

Product governance continues after contract award. Regulatory approval, batch quality, labelling, storage conditions, distribution integrity, complaint handling and recall capacity are part of the legal architecture of implementation. A standardised global formulation does not guarantee that every locally supplied batch, package or instruction will be suitable. Nor does classification as an essential medicine displace national regulatory judgment. Quality failures would be especially damaging because they could erode trust in antenatal services beyond MMS itself. The programme should therefore integrate regulatory data and field complaints into the same adaptive review that evaluates coverage and outcomes.

5. MMS within Indonesia's Structural Nutrition Problem

Indonesia-specific research reinforces the need for a systems approach. Evidence has linked child stunting to water, sanitation and hygiene conditions alongside maternal and household factors. Broader review identifies a dense set of determinants, including maternal nutrition, birth outcomes, infant feeding, infection, food insecurity, care practices, socioeconomic position and service access. The legal risk is not that MMS lacks value, but that a countable tablet becomes the visible substitute for harder obligations across sectors.

This is the problem of tabletisation: translating a structural condition into a commodity-centred programme whose success is measured mainly by distribution. Tabletisation can occur even when policymakers acknowledge multiple causes. Budgets, dashboards and accountability meetings may still privilege what can be purchased and counted. Presidential Regulation No 72 of 2021 provides a legal counterweight because its holistic design can be used to insist that supplementation remains one component of a broader duty.

Street-Level Governance: Where The Standard Meets The Woman

1. Discretion in Routine Maternal Care

National adoption does not determine how MMS is experienced. Frontline workers interpret eligibility, explain benefits, respond to nausea or other complaints, reconcile new and old stock, record adherence and decide when to refer. Street-level bureaucracy theory shows that public workers inevitably exercise discretion under conditions of limited time, resources and organisational pressure. A formally uniform programme will therefore generate locally variable practices.

Public accountability at street level cannot be secured by central rules alone because workers answer simultaneously to professional norms, supervisors, indicators and service users. If coverage targets dominate, counselling may be compressed into tablet handover. If stock is unstable, workers may ration or switch products without clear communication. If reporting systems penalise discontinuation, side effects may be under-recorded.

Research on nurses as street-level bureaucrats demonstrates how frontline personnel can remain committed to public goals while experiencing frustration, constraint and moral burden. MMS implementation should therefore avoid treating non-compliance as a failure of motivation by either the woman or the worker. Administrative design must provide time, training, stock reliability, escalation channels and discretion bounded by rights.

2. Organisational Power and the Production of Compliance

Implementation organisations are sites of power, not neutral delivery pipes. Rules, performance systems, hierarchy and professional status affect whose knowledge counts and which problems become visible. A pregnant woman who reports discomfort or distrust may encounter a system organised to achieve coverage rather than to negotiate informed adherence. Local managers may also lack authority to alter supply or reporting rules even when they recognise implementation problems.

Experiences with iron supplementation show that women's understandings of anaemia, perceived benefits, side effects, family influence, access and quality of counselling shape use. MMS adds nutrients but does not remove these behavioural and relational conditions. Communication must avoid implying that a woman is individually responsible for fetal growth despite poverty, food prices, work demands or inadequate services.

3. The Implementation Gap and Adaptive Learning

Classic implementation analysis shows how policy objectives are transformed by long chains of decisions and dependencies. Indonesia's geography, decentralisation and service diversity lengthen that chain. A credible research-context implementation should therefore publish not only aggregate coverage and outcome trends but also supply interruptions, discontinuation, complaints, regional inequities, counselling quality and reasons for non-use.

Digital nutrition information systems can support learning, but data architecture should not be confused with knowledge. Indicators select what is seen. Data quality may vary with workload and incentives. Privacy, access control and secondary use require governance. Most importantly, feedback must have institutional consequences: the ability to revise communication, procurement, dosage instructions, target populations or programme pace.

Equity analysis should move below the national average. Island geography, remoteness, disability, language, migration status, adolescent pregnancy, poverty and uneven facility capacity can affect both access and the quality of counselling. A programme can report high

aggregate coverage while reproducing sharp differences in continuity and informed use. Distribution data should therefore be disaggregated where lawful and methodologically sound, and qualitative complaints should be treated as evidence rather than anecdote. Equity is not an additional outcome after scale; it is a condition of lawful scale.

Legal And Ethical Implications: Accountability Without Command

1. The State Remains the Primary Duty Bearer

The right to health in the International Covenant on Economic, Social and Cultural Rights requires states to take steps towards the prevention and treatment of disease and the creation of conditions assuring medical service. External support can help discharge that duty but cannot transfer it. The state remains responsible for deciding whether MMS is appropriate, ensuring quality and equitable access, and maintaining the rest of the maternal health system.

The availability, accessibility, acceptability and quality framework provides a practical rights test. Availability concerns reliable stocks and trained personnel. Accessibility includes geographic, economic, informational and non-discriminatory access. Acceptability requires culturally and ethically respectful counselling and attention to autonomy. Quality includes product standards, storage, prescribing or dispensing practice, adverse-event response and accurate information.

Women's equality rights reinforce the need to resist maternal instrumentalisation. CEDAW requires appropriate services in connection with pregnancy and adequate nutrition during pregnancy and lactation. The Convention on the Rights of the Child likewise recognises maternal and child health, nutrition, clean drinking water and preventive care within a connected set of obligations. A rights-based programme must therefore serve the woman and the child without making either a justification for neglecting the other.

2. Law as a Determinant of Health and a Constraint on Governance

Law can improve health by creating entitlements, financing institutions, regulating products and structuring accountability; it can also generate exclusion, fragmentation or harmful incentives. In the MMS corridor, domestic law should perform two functions. It should enable timely implementation where evidence and need justify action. It should also slow the process at specific points where justification, participation, procurement integrity or rights protection require scrutiny.

Proportionality offers a useful discipline. Adoption should pursue a legitimate health aim, use means reasonably connected to that aim, avoid unnecessary burdens on autonomy or competing services, and maintain a defensible balance between expected benefit, uncertainty, cost and opportunity cost. Precaution should not be invoked mechanically either for or against MMS. Where potential benefit is meaningful and serious harm is unlikely, uncertainty can justify monitored action. Where uncertainty concerns system displacement, dependence or inequitable implementation, the appropriate response is not paralysis but safeguards, staged commitment and reversible design.

Global governance institutions derive legitimacy not simply from beneficial outcomes but from acceptable procedures, epistemic quality and capacity for correction. Accountability likewise requires standards against which conduct is judged, information about performance and consequences or responses when standards are not met. These conditions apply differently to WHO, donors, ministries, regulators, suppliers and health facilities, but no influential node should be wholly outside them.

3. Six Safeguards for the Normative-Material Corridor

First, domestic legal translation. International guidance should be converted into a publicly identifiable domestic instrument or decision that states the legal authority, target population, product standard, place within antenatal care, responsible institutions and review date. Translation prevents the guideline from operating as an uncodified external command.

Secondly, public reason-giving. The government should publish a proportionate evidence-to-decision account addressing benefits, uncertainty, local anaemia and dietary conditions, feasibility, equity, costs, alternatives and the interpretation of the research condition. The purpose is not to reproduce a global evidence review, but to show why the intervention is justified in Indonesia and how contrary considerations were handled.

Thirdly, financial and relational transparency. Donor agreements, matching formulas, product or supplier relationships, technical-assistance roles, data arrangements and exit assumptions should be disclosed at a level that permits public understanding and audit. Transparency must illuminate influence, not merely publish a total funding figure.

Fourthly, adaptive authorisation. Scale-up should be linked to pre-specified review points and decision rules. Data on coverage, birth outcomes, anaemia, side effects, adherence, equity, supply reliability, counselling and opportunity costs should be capable of changing the programme. Shared responsibility for structural injustice requires institutions to revise the conditions they help produce rather than assigning blame only to individuals at the endpoint.

Fifthly, a systems obligation. Public authorities should demonstrate that MMS expenditure and performance targets do not displace necessary measures addressing food security, primary care, infection, water and sanitation, social protection and respectful maternity services. Public-health justice is concerned with the distribution of multiple dimensions of well-being, not only a single biomedical outcome.

Sixthly, a distributed accountability matrix. Each actor should be linked to its function, standard, information duty, review mechanism and remedy. WHO should explain and update the evidentiary basis of its recommendation. Donors should disclose conditions and plan sustainable exit. National authorities should justify, regulate and finance. Procurement bodies should protect integrity and resilience. Regulators should assure quality. Facilities should deliver respectful care and complaints mechanisms. Trust is strengthened not by demanding confidence in institutions, but by making institutions intelligently answerable.

4. Experimentation, Review and Remedies

A research-context recommendation is compatible with phased and adaptive governance only if learning can alter the common framework. Experimentalist models emphasise provisional goals, local discretion, reporting, peer comparison and periodic revision. Applied to MMS, this means local implementation should generate reasons for national and global adjustment, not merely evidence of compliance with a fixed model.

Participation should also be designed around the decisions participants can actually influence. Consultation after the formulation, financing model and procurement specification have been settled is informational rather than deliberative. Pregnant women, professional associations, local governments, disability and women's rights organisations, nutrition experts, pharmacists and domestic producers will not hold identical interests. A legitimate process does not require consensus. It requires timely access to information, an intelligible

account of trade-offs, reasoned responses to material objections and representation of groups that bear implementation burdens without controlling the agenda.

Institutional design should specify who can demand an explanation, who can investigate failure and what happens next. Remedies need not always be judicial. They may include administrative complaint, procurement review, disclosure orders, audit findings, corrective action, product recall, protocol revision, budget reallocation or compensation where legal standards are breached. Judicial review remains an important backstop where governmental authority, procedure or rights are violated.

5. Maternal Autonomy and Structural Non-Blame

Public health ethics requires attention to benefits, burdens, liberty, justice, transparency and the moral quality of implementation. A supplementation programme should offer clear information about purpose, expected benefit, uncertainty, administration and side effects. Routine recommendation may be appropriate, but it should not collapse into pressure or stigma.

Structural violence describes how political and economic arrangements become embodied as unequal risk. Maternal undernutrition is shaped by such arrangements. Public communication should therefore avoid representing MMS as a test of whether a woman is responsible, compliant or caring. The state cannot distribute a tablet and then attribute an adverse outcome to individual behaviour while leaving structural conditions unaddressed.

Fair health policy must consider both health outcomes and the fair distribution of the social conditions that produce them. MMS may be part of that policy because a feasible intervention should not be withheld while structural reform proceeds. The legal requirement is complementarity: rapid supplementation and slower structural measures must be governed as connected obligations rather than competing narratives.

CONCLUSION

The global expansion of MMS demonstrates how contemporary health governance operates beyond the conventional boundaries of law. WHO recommendations, essential medicine recognition, expert networks, catalytic finance, product standards, procurement support, and monitoring mechanisms do not constitute a single binding command. Instead, they form a normative-material corridor that guides policy by making one option more legitimate and feasible than others.

Indonesia's 2026 program demonstrates both the potential value and constitutional complexity of this arrangement. External cooperation can accelerate access, strengthen supply systems, and support domestic production. Indonesia also retains meaningful agency through political commitment, co-financing, regulation, and implementation. However, the state exercises this agency within a policy environment in which evidence, terminology, product configurations, and enabling resources are organized through transnational processes. Formal consent is therefore necessary but insufficient as a complete account of legitimacy.

Medical law should respond by tracing power across institutional boundaries while maintaining a distinction between influence and legal responsibility. The state remains the primary duty bearer toward women and children. WHO, donors, consortia, and suppliers should be accountable for the specific functions through which they influence policy formation and implementation. Domestic legal translation, reason-giving, transparency,

adaptive review, systems obligations, and distributed accountability can strengthen the legitimacy of the corridor without undermining its capacity to deliver beneficial interventions.

This analysis is necessarily limited by its documentary methodology. Future research should examine the legal structure of program agreements, the development of Indonesian procurement specifications, interactions between central and local budgets, the experiences of frontline workers and pregnant women, the role of domestic manufacturers, and whether implementation evidence influences national or WHO policy. Comparative research across countries could further assess whether the corridor operates differently in contexts with stronger independent evidence assessment, procurement capacity, and domestic manufacturing capabilities.

The central lesson extends beyond maternal nutrition. Global health increasingly operates through standards connected to resources. When such combinations influence what states can realistically choose, public law must examine not only whether a rule is formally binding, but also how a policy pathway becomes difficult to reject, who benefits, who bears the risks, and whether those shaping the pathway remain open to contestation, correction, and remedy.

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